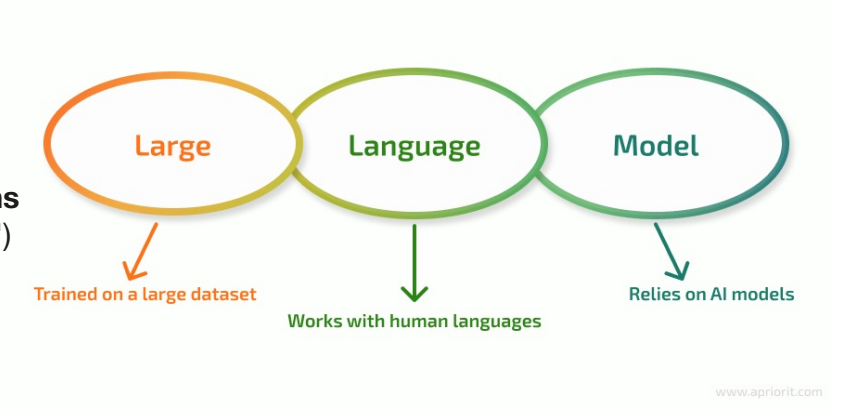
LLM

- LLM = **Large Language Model**
- **Neural networks**, AI trained to **understand** & **generate** human language
- NOT a database — it **predicts**, doesn't look up
- "Large" = billions of parameters (tunable numerical weights)
- Examples: **GPT-4, Claude**, Gemini



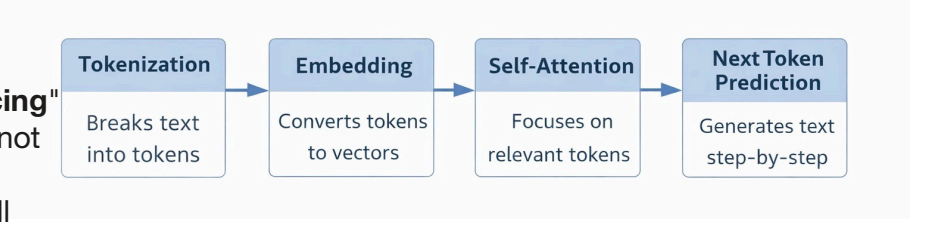
How do they work ?

1. Tokenization & Embeddings

- **Text** broken into **chunks** called **tokens** ("unhappiness" → "un" + "happiness")
- Each **token** → converted to a **vector** (numbers capturing meaning)

2. Self-Attention

- Every **token** looks at every **other token** & decides **relevance**
- Ex: "**What is load balancing**" → "**load**" pays **attention** to "**balancing**" → understands it's a **tech term**, not physical load
- Result → **model understands** full **context** before generating anything



3. Next token prediction

- Outputs **probability** for every possible **next word** → **picks highest** → **appends** → **repeats**
- Input is "**What is load balancing**" → model sees a definition pattern → picks **"a"** with **38% probability**
  - Now input is "What is load balancing a" → "a" signals a noun is expected next → **picks "technique" with 32%**
  - Now input is "...a technique" → "technique" signals something describing what it does → **picks "used" with 41%**
  - Now input is "...technique used" → "used to" is a standard English pattern → **picks "to" with 72%**
  - Now input is "...used to" → model trained on millions of tech docs knows "distribute" follows load balancing context → **picks "distribute" with 45%**
- Final output: "**Load balancing is a technique used to distribute traffic across servers**"

